Particle Physics

A Layperson's Hypothesis – Preliminary Notes

Author: Andrew Robus
Date: November 13, 2026
License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Hypotheses

- There is only one elementary particle: the energy quantum.
- Our Universe is a black hole.
- Universe = E (representing general relativity in a purely energy-based manner, since everything is energy).

Particle Physics

1. Introduction

In this hypothesis, we start from the following assumptions:

- Universe = E and E = strings.
- General relativity is fully preserved.
- All formulas and insights regarding how photons can be converted into electrons, quarks, and ultimately all other particles remain valid.
- We integrate modern particle physics into Einstein's light quantum hypothesis, showing why a particle is simultaneously a particle and a wave while consisting of individual strings.

Einstein lived from 1879 to 1955. Particle physics was still relatively manageable back then.

Historical classification

Particle	Discovery or classification
Electron	1897
Light quantum / Photon	Light quantum hypothesis, 1905
Proton	Nuclear model, 1911; proton designation, 1920
Neutrino	Postulated in 1930
Neutron	1932
Positron	1932
Muon	1936
Neutrino	Experimentally detected in 1956, after Einstein's death

2. The Energy Quantum

A string is the fundamental building block of the universe, and every single elementary particle (excluding the gluon) consists of a single string.

It is a thread-like piece of energy. Whether this thread-like appearance is its true nature or is caused by length contraction triggered by motion at the speed of light (c) remains unclear.

Whether this energy quantum flies forward or backward (likely its helicity) determines whether the energy quantum is positive or negative.

For example: Two photons have the same spin and flight direction. If one photon flies with its "head" forward and the other with its "foot" forward, their helicity and spin projection are opposite, even though both photons are identical energy quanta. This is why an energy quantum is its own antiparticle.

As long as this energy quantum encounters nothing, it is invisible because it emits no energy. Light only becomes visible when energy encounters other energy.

When this energy quantum encounters a stable particle (stable particles always consist of positive and negative parts), energy flows from positive to negative, or a short circuit occurs due to identical polarity. This is a detail question for physicists.

The result: Light shines because energies are exchanged, interact, or transform.

Energy, wavelength and geometry

Its wavelength could correspond to its geometric length, thereby establishing a connection to its amount of energy.

According to the familiar relationship $E = h \times f$:

- The higher the frequency, the higher the energy.
- The higher the frequency, the shorter the wavelength.

Applied to this model, this would mean that a shorter string represents higher energy.

The amount of energy could additionally or primarily depend on its diameter or cross-sectional area (or width).

A more energetic string could accordingly be shorter but simultaneously thicker (wider). The energy would then be more strongly concentrated in a smaller spatial range.

Assuming constant energy density, a larger cross-section could accommodate more energy. Alternatively, the energy density itself could vary with the diameter.

Spatial length and diameter would thus be two distinct geometric properties of the energy quantum that together could determine its energy content.

Whether the energy is actually distributed across these geometric properties and how length, diameter, and energy density precisely relate to one another remains unresolved.

AI note: We treat your baseline assumption—that rest energy and kinetic energy are merely different manifestations of the same energy—as the starting point of the thought experiment.

3. Six Photons Compared

We calculate the energy using the relationship:

$E = (h \times c) / \lambda$

Here, we initially set the wavelength as the geometric length of the string.

Range	Example	Wavelength (assumed string length)	Energy
Low	VHF, 1 MHz	300 m	0.00000000414 eV
Low	VHF, 100 MHz	3 m	0.000000414 eV
Medium	Red light	700 nm	1.77 eV
Medium	Green light	500 nm	2.48 eV
High	X-rays	0.1 nm	12,400 eV
Very high	Gamma rays	1 pm	1,240,000 eV

4. Can We Calculate the Thickness from This?

Now we test your geometric concept.

We provisionally assume:

- The string is cylindrical.
- Its length corresponds to the wavelength.
- Its energy is uniformly distributed over its volume.
- The energy density remains identical for all photons.

Then:

Energy = energy density × cross-sectional area × length

For a circular cross-section:

$E = u_E \times (\pi \times d^2 / 4) \times L$

Now we substitute the assumption that the string length L equals the wavelength λ and the familiar relationship $E = (h \times c) / \lambda$.

This yields:

$d^2 = (4 \times h \times c) / (\pi \times u_E \times \lambda^2)$

And thus:

d is proportional to 1 / λ, which is proportional to E.

Interesting result within the model

If we assume a constant energy density, the diameter (or width) would be directly proportional to the photon energy.

5. How Much Would the Diameters Differ?

We take the green photon at 500 nm as a reference and set its hypothetical diameter to an arbitrary unit.

Photon	Relative diameter
VHF 1 MHz	0.00000000167
VHF 100 MHz	0.000000167
Red 700 nm	0.714
Green 500 nm	1
X-ray 0.1 nm	5,000
Gamma 1 pm	500,000

These values are not measured photon diameters, but follow exclusively from our model assumptions.

What Does This Mean?

This yields an interesting geometric picture:

- Radio photons would be extremely long and thin.
- Visible photons would be significantly shorter and thicker (wider).
- X-ray photons would be very short and considerably thicker.
- Gamma photons would be extremely short and very thick.

This would give us a possible explanation for why higher-energy photons possess a smaller length scale despite containing more energy.

Additional note: If width instead of diameter is responsible for the energy content, then due to motion at the speed of light parallel to the direction of flight, the string would become a point regardless of how much energy it contains. In that case, a 1 pm gamma ray would not necessarily leave a thicker point on the detector than lower-energy photons.

The diameter model would also work if the resulting light spot were generated precisely at the center.

6. Photon

A photon is not a particle. A photon describes what we measure when an energy quantum encounters a stable particle.

The energy quantum is positive or negative, depending on its direction of flight. Alternatively, the front could be positive and the back negative because it is being pulled.

Stable particles always consist of positive and negative components.

The energy quantum either shines when energy can flow between positive and negative, or a short circuit occurs between two identical poles, or both possibilities apply.

This is a detail question for physicists.

The Electron

7. Possibility A

A single string is curved so sharply that an arc forms between its positive and negative ends, or the string even forms a closed circle without an arc.

It is like a dog chasing its tail.

The energy now flows in a circle in place, forming a stable particle with rest mass.

8. Possibility B

The energy quantum remains a normal string. It has been forced into an intrinsic spin.

The same energy continues to move at the speed of light (c). The motion now simply takes place over a very limited area.

Whether this is an intrinsic spin momentum, a rotation, or an orbital path is a detail question.

By moving at the speed of light "in place", it possesses "rest mass" and becomes fixed.

Rest mass is thus the same kinetic energy—or simply energy—as that of a massless photon.

Quarks and the Proton

Quarks also consist of the energy quantum.

However, we need the proton to understand them more easily. It consists of two up quarks and one down quark (two positive, one negative; this depends on whether the string flies forward or backward).

9. Possibility A – Ring Structure

A quark is a circular energy quantum chasing its tail, like the electron from Possibility A, though the quantum properties are different.

The quantum numbers and descriptions from the Standard Model of particle physics can be adopted, as they already describe the properties.

The arrangement is:

Up – Down – Up

There are three rings rotating parallel to each other, attracting each other or even interlocking, like a chain of rings.

Plus and minus attract.

Therefore, particles consisting purely of negatively or purely positively rotating strings can only exist temporarily. They repel each other and decay into the nearest possible stable state, defaulting to a photon. Neutrons and neutrinos are left aside here.

The color charge arises either between the parallel rotating rings or, in the ring chain, in the region where the inner surfaces of an up quark and a down quark overlap.

Interlocking mechanism

Probably due to polarization and gravitational/electromagnetic interaction, the quarks are "interlocked" with each other.

If you pull them apart, this interlocked region stretches.

Perhaps this can be imagined like two rubber bands welded together: as you stretch them, the area becomes thinner and resistance increases until it tears again at the welded seam.

10. Possibility B – String Chain

A quark, like any other elementary particle, is simply an energy quantum (string).

Depending on energy content, helicity, and spin, it appears as an up quark, a down quark, or any other elementary particle, aside from the gluon.

I cannot say anything about bosons yet.

Here, the three quarks form a chain, with the down quark in the middle acting as the connecting piece.

The spin is divided into thirds because they now form a continuous chain and the spin is distributed across the entire structure.

Color charge works here just like in Possibility A, except that it sits where the up quarks dock onto the down quark.

Conclusion

The exact motion patterns of the strings are not yet clarified and are likely represented in a heavily simplified manner here.

Strings could also consist of a piece of electric wave and a magnetic wave instead of positive and negative.

However, it is possible that they also possess positive and negative flight directions. What standard physics currently refers to as positive and negative in this context, I would then designate as left and right. The principles would be almost identical.

Unrefined notes on this are on the following pages.

It is also conceivable that the quanta move in a wave-like manner, though I consider this unlikely.

The only intrinsically stable particle is the fundamental building block: the energy quantum.

Alternatively, since a string can break off and vary in width or diameter—and can also break apart, as in photon splitting—perhaps an energy quantum is a quantum of a string rather than every photon.

This is something for quantum physicists to decide.

Additional Notes

11. The Calculation Using Your Definition

Your photon is a string of length L.

It flies past the detector at speed c; the front and back arrive successively.

This takes a time interval of Δt = L / c, and in your view, this time duration is an oscillation period.

From this follows the oscillation frequency:

f = c / L

In standard physics, this is the relationship between frequency and wavelength:

λ = c / f

Your length L thus corresponds to the wavelength.

Comparing the energy relationships

Now we compare this with your energy rule:

E = λ_E × L

Here, energy increases with length.

We calibrate it to green light: 500 nm = 2.48 eV.

The measured relationship is:

E = h × f = (h × c) / λ

Light	Length L	Measured energy	Energy according to E proportional to L
X-rays	0.1 nm	12,400 eV	0.0005 eV
Green	500 nm	2.48 eV	2.48 eV
VHF Radio	3 m	0.0000004 eV	Approx. 15 million eV

Your radio photon would thus have the energy of a gamma quantum, and the X-ray photon almost none.

Nature does it the other way around: shorter means more energetic.

If you change your rule to:

E = h × f = (h × c) / L

the entire table fits.

Then the energy is not uniformly distributed over the string, but is concentrated more strongly in shorter strings.

12. Matter Shrinkage and the Hubble Constant

Static model, with H_o = 70 km/s/Mpc

First, the unit:

70 km/s per Mpc yields:

$H_0 \approx 2.27 \times 10^{-18}$ per second

or:

7.2×10^{-11} per year

This means: For your shrinking matter to mimic the observed expansion, every length within matter would have to become shorter by about 7 ten-billionths per year.

The energy per quantum (E proportional to 1/L) increases in the same proportion.

Shrinkage per year

Object	Shrinkage per year
Human (1.8 m)	0.13 nm, about one atomic diameter
Human over 80 years	About 10 nm
Earth (radius)	0.46 mm
Earth–Sun distance	About 11 m

This is the figure for your hypothesis: You never notice it because the scale, atomic clock, and you yourself shrink in the same proportion.

Redshift

A galaxy with z = 1 emitted its light when all matter was twice as large as it is today.

At the microwave background (z ≈ 1100), it was around 1,100 times larger.

13. Variable Shrinkage Rate – H as a Function of Time

H_0 is only the current value of H(t).

In general, in your picture:

- The relative shrinkage rate is H(t).
- The size of matter is proportional to 1/a(t).

With the standard model (flat, $\Omega_m \approx 0.3$, $\Omega\Lambda \approx 0.7$), the relationship is:

$H(z) = H_0 \times \sqrt{[0.3 \times (1 + z)^3 + 0.7]}$

Shrinkage rate at different redshifts

Redshift z	Shrinkage rate per year
0 (today)	7.2×10^{-11}
0.5	9.4×10^{-11}
1	1.3×10^{-10}
2	2.1×10^{-10}
10	1.4×10^{-9}

In the past, matter therefore shrank faster; today it shrinks more slowly.

Your thought that compression proceeds with acceleration later corresponds here to a shrinkage rate approaching a constant value, about 6×10^{-11} per year.

Then everything shrinks exponentially.

A possible test of the hypothesis

Therein lies an opportunity:

The curve used here comes from the standard model.

In your hypothesis, it should follow from the dynamics of the strings—that is, from how energy flows toward the center.

If you formulate this as a formula, it could be compared with measured H(z) values from supernovae.

This turns the idea into a testable prediction.

One note: Measurements of H_0 itself are currently debated between roughly 67 and 73, but this does not change the order of magnitude.